

Toby Tran

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EDUCATION

University of South Australia

Bachelor of Information Technology (Software Development)

Mawson Lakes, SA

Mar. 2023 – Nov. 2025

- **GPA:** 6.17 / 7.0
- **Honours/Awards:** 2023-25 University Merit Award(s)
- **Relevant Coursework:** Data Structures Essentials, Object-Oriented Programming (OOP), Design Patterns with C++, Web Technology, Agile Development and Governance, Operating Systems and Tool Chains.

PROJECTS

Mission Planner | *FastAPI, PyWebView, ReactJS, TailwindCSS, CesiumJS, DBScan*

Jul. 2025 – Nov. 2025

- Worked in a **3-person team** to develop a full-stack desktop application for an orbital debris removal startup (**Paladin Space**) to identify high-risk collision targets for commercial satellites, saving significant mission costs.
- Applied a **layered backend architecture** that splits the system into multiple domains, combined with design patterns (*Strategy, Adapter, Factory*) for easier maintenance and future extensions.
- Implemented **10+ asynchronous FastAPI REST API endpoints** for a non-blocking data processing experience, which prevents the application from freezing while the data is being processed/synchronised.
- Led a backend data processing pipeline utilising HTTPX to handle concurrent bulk downloads of **100,000+ CSV records**, serialising and persisting data locally for low-latency retrieval and cross-system data compatibility.
- Built an **automated synchronising pipeline** that tracks dedicated logs to validate data freshness against a custom threshold and trigger a sync event if prompted, supporting near real-time fidelity for mission planning.
- Integrated a 3D globe using CesiumJS web components to visualise the processed data, providing users with a better demonstration of the impact of the collision targets.

SA Expiation Dashboard | *ASP.NET Core Web API, LINQ, ReactJS, Bootstrap, D3.js*

Jul. 2024 – Nov. 2024

- Developed a ReactJS dashboard with up to **6 different search/filter controls** to identify **optimal Mobile Phone Detection Camera (MPDC) set-up locations** from expiation data.
- Integrated secure authentication and login features with **SHA-256** password hashing to prevent unauthorised access to sensitive records.
- Implemented **6 WebAPI REST API endpoints** to query location-based expiation records for strategic offense analytics, using LINQ (EF Core) queries to safely fetch MS-SQL data.
- Designed a SAPOL-branded user interface with clear visual hierarchy using consistent Bootstrap styles to facilitate operator workflow.
- Visualised MPDC location insights from search results via **interactive D3.js graphs** incorporated within a detailed report page to help operators easily spot hotspots and prioritise deployment areas.

Towny Title | *Spigot API, Java, Maven, Git*

Oct. 2017

- Developed an open-source Minecraft server plugin to enhance player navigation and territory awareness for the Towny framework.
- Implemented **real-time player location tracking and event-driven triggers** to render dynamic on-screen titles and warnings with minimal server latency.
- Implemented cross-version compatibility across **6 major Minecraft releases** (v1.7 to v1.12) to support diverse server environments.
- Published the plugin to SpigotMC, reaching **1,200+ downloads** and helping multiple servers enrich their territory simulation experiences.

TECHNICAL SKILLS

Languages: Java, Python, C#, C/C++, SQL (MySQL), JavaScript, HTML/CSS

Frameworks: ASP.NET Core (MVC, WebAPI), React, TailwindCSS, Node.js, FastAPI, Framer Motion

Developer Tools: Git, Docker, VS Code, Visual Studio, IntelliJ, Eclipse, Jira Atlassian (Scrum-based)